# Introduction to Excel in Data Analytics

## Introduction to Excel:

- Microsoft Excel is the ***Spreadsheet program that allows users to store, organize and analyze Information***

## Excel Interface:

- **Ribbon:**
    - Ribbon contains ***tabs***, each with *multiple groups of commands related to a specific task*
    - *Show Ribbon* options:
        - ***Auto-hide Ribbon / Full-Screen Mode***
        - ***Show Tabs***
        - ***Show Tabs and Commands / Always Show Ribbon***
- **Worksheet:**
    - Worksheet is a ***Grid of Cells organized into Rows and Columns*** where we can *Enter & Manipulate Data*
    - It holding Numerical data, Text, Formulas or Functions
    - To Duplicate the Dataset in New workbook — In ***Sheet1 —> Move or Copy —>To book —> (new book) & Create a Copy***
- **Workbook:**
    - It is ***File*** contains ***one or more Worksheets*** which users to organize & analyze the data
- **Rows:**
    - Rows are Horizontal collections of cells in Excel (1,2,3, 4...)
    - Total number of Rows on a worksheet - ***10,48,576***

- **Columns:**
    - Columns are Vertical collections of cells in Excel (A to Z, AA to ZZ…)
    - Total number of Columns on a worksheet - ***16,384***

- **Cell:**
    - The ***Intersection of a Row and a Column is called a Cell***

- **Cell Reference:**
    - Each Cell has a ***Unique Address***
    - It is Identified by a Combination of Column letter and Row number (***A1, B2***)

- **Table:**
    - It is a ***Collection of Data organized in Rows and Columns***

- **Range:**
    - Range refers to a ***Selection of Multiple Cells*** in a Excel (Group of Adjacent Cells)
    - It usually defined by **“Starting Cell”** and **“Ending Cell”** separated by **Colon - *[A1:C5]***
    - It may be *a Single Cell (A1) / a Row (A1:C1) / a Column (A1:A3) / a Range (A1:C5)*

- **Inserting Rows:**
  - ***Method 1: Ribbon Command***
    - *"Home" —> "Cells" —> "Insert" —> "Insert Sheet Rows"*
  - ***Method 2: Right Click option***
    - Right click on row number —> ***"Insert"*** option from context menu
  - ***Method 3: Shortcut***
    - Insert Rows
      - ***Alt + I + R*** (or)
      - ***Ctrl + Shift + '+'***

- **Inserting Columns:**
  - ***Method 1: Ribbon Command***
    - *"Home" —> "Cells" —> "Insert" —> "Insert Sheet Columns"*
  - ***Method 2: Right Click option***
    - Right click on column letter —> ***"Insert"*** option from context menu
  - ***Method 3: Shortcut***
    - Insert Columns
      - ***Alt + I + C*** (or)
      - ***Ctrl + Shift + '+'***

- **Deleting Rows:**
  - ***Method 1: Ribbon Command***
    - *"Home" —> "Cells" —> "Delete" —> "Delete Sheet Rows"*
  - ***Method 2: Right Click option***
    - Right click on row number —> ***"Delete"*** option from context menu
  - ***Method 3: Shortcut***
    - ***Ctrl + '-'*** —> Delete Rows

- **Deleting Columns:**
    - ***Method 1: Ribbon Command***
        - *"Home" —> "Cells" —> "Delete" —> "Delete Sheet Columns"*
    - ***Method 2: Right Click option***
        - Right click on column letter —> ***"Delete"*** option from context menu
    - ***Method 3: Shortcut***
        - ***Ctrl + '-'*** —> Delete Columns

## Google Spreadsheet:

Online Version of MS Excel same as Local MS Excel

## Importing Data from Delimited Text files:

**Step 1:** Open ***Notepad*** and Enter your data with ***Comma (or) Space as delimiter***

**Step 2:** Save As the file with .***CSV format*** *(Comma Separated Value)* → ***"Filename.csv"***

**Step 3:** Data Import Process -

- Click on ***"Data" → "Get External Data" → "From Text" → "Delimited Text file location"*** → Click ***"Import"***

**Step 4:** Text Import Wizard:

- select ***"Delimited"***
- select ***"My data has headers"***
- Choose the Delimiter *(Comma / Space)*

**Step 5:** Configure '***Datatypes***' (Optional)

**Step 6:** Import Location - Excel ***Existing Worksheet / New Worksheet***

or

**Step 7:** Click on ***"Data" → "From Text / CSV"*** → choose ***".CSV file" → "Import" → "LOAD"***

## Table Creation in Excel:

- Select the Cells for Table Creation
- Table Creation Options:
    - ***Insert —> Table —> Create Table***
    - Shortcut —> ***Ctrl + T***
    - ***Home —> Format as Table***
- In **Create Table** option, select ***cell reference*** for Table Creation & Mark ***"My table has headers"***
- Table was created
- Table Naming — ***Ctrl +T —> Create Table Design —> Rename the "Table Name"***
- To Add Rows in that Table - Press ***"Tab"*** button

## Transpose the Column Headers:

- **Step 1:** Copy the Column Headers and Paste it in another sheet
- **Step 2:** In *right bottom corner of the Paste Column Headers —> click* ***"Ctrl"*** *—> select* ***"Transpose"*** *option*
- Shortcut —> ***Alt+e+s*** - (Paste Special —> Transpose)

## To Access the Data using Excel Formulas:

- Example:
    - ***=SUM(TABLE_NAME[COLUMN])***

## Benefits of using Excel Tables:

- ***Structured Formatting*** - Pre-Defined Styles for formatting the Table
- ***Dynamic Range*** - Add New Data easily